Chapter 8

Homological algebra

In this chapter we develop some basic homological algebra tools, which are necessary to study the fine structure of representation categories.

8.1. Projective and injective modules

Let A be an algebra, and let P be a left A-module.

Theorem 8.1.1. *The following properties of P are equivalent:*

(i) If $\alpha : M \to N$ is a surjective morphism and $\nu : P \to N$ is any morphism, then there exists a morphism $\mu : P \to M$ such that $\alpha \circ \mu = \nu$.

(ii) Any surjective morphism $\alpha : M \to P$ splits; i.e., there exists $\mu : P \to M$ such that $\alpha \circ \mu = \mathrm{id}$.

(iii) There exists another A-module Q such that $P \oplus Q$ is a free A-module, i.e., a direct sum of copies of A.

(iv) The functor $\mathrm{Hom}_A(P, ?)$ on the category of A-modules is exact.

Proof. To prove that (i) implies (ii), take $N = P$. To prove that (ii) implies (iii), take M to be free (this can always be done since any module is a quotient of a free module). To prove that (iii) implies (iv), note that the functor $\mathrm{Hom}_A(P, ?)$ is exact if P is free (as